Failure as a Theorem: Certified Error Objects, Fidelity Auditing, and the Instrumented Last Mile in an LCF Kernel for Bourbaki's Theory of Sets

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Abstract

Proof assistants certify success: a theorem either checks or it does not. We present a system in which *failure* is certified with the same rigor. Working in an LCF-style kernel implemented directly over Bourbaki's own formal system — the τ operator and its abbreviated assemblies, a formalism no prior mechanization inhabits — we develop (i) *certified failure objects*: every dead end carries a kernel-checked certificate (e.g. a derivation of absurdity from a residual hypothesis) and a *computed* blast radius; (ii) an operational trichotomy classifying every open hypothesis as dischargeable, refutable, or independent — the first two verdicts kernel objects, the third a default whose epistemic status we state explicitly; (iii) *debt measurement*: a theorem advertised as hypothesis-free is measured to consume 53 axioms of ad-hoc theories invisible to its statement, motivating a sound accounting of what a derivation actually assumes; and (iv) *mechanized fidelity auditing* against the source text at page granularity, under which two axiom transcription defects were detected by *deriving a contradiction*, repaired by restoring the book's form, and proven fixed by the impossibility of re-deriving it. A Weisfeiler-Leman similarity scan reduces the detection of one such incoherence from a 95-minute two-agent investigation to a query of a few seconds. We report an instrumented week of operation — three axiom defects found and repaired, a vacuous theorem revoked and later rehabilitated by discharge, a 3,909-test suite — and argue that the resulting corpus of certified failures, unavailable from any existing proof corpus, is the missing training substrate for AI systems intended to *create* theories rather than merely search for proofs.

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1 Introduction

Proof assistants are engines of certainty: a theorem checks or it does not, and libraries accumulate the successes. Everything else — the failed attempts, the routes that dead-ended, the hypothesis that turned out refutable, the axiom that was mistranscribed from the book — is discarded, surviving at best as commit messages and folklore. Early instrumentation work showed this discarded material can be captured [35], and the current generation of proving agents consumes failures as training negatives or repair prompts [41, 8]. But in every system we know, the failure itself remains an *unverified* artifact: a compiler message, a model's own diagnosis, a judged trajectory. Nothing guarantees that the recorded wall was real, that its declared cause is the actual cause, or that the search it forbids is actually hopeless.

This paper takes the opposite stance: **failure can be certified with the same rigor as success**. In our system a dead end is an object carrying a kernel-checked certificate — for instance a derivation of absurdity from the residual hypothesis that killed the route — together with a *computed* scope stating exactly which goals it condemns. An infidelity to the source text is not an estimated alignment score but a derived contradiction. What a theorem silently assumes is not read off its hypothesis count but measured over its derivation. And the corpus of such objects demonstrably steers subsequent search: routes certified dead are never paid for twice, and a revoked theorem, kept with its certificate, can later be rehabilitated unchanged when its theory is repaired.

The setting sharpens the experiment. We work in an LCF-style kernel implemented over Bourbaki's *own* formal system — the τ-operator, its abbreviated assemblies, and the body of criteria Bourbaki *proves* about demonstrations instead of positing as rules — a formalism that, to our knowledge, no prior mechanization inhabits (Section 2). What the kernel implements is Bourbaki's notion of a *theory*: any name carrying a finite set of explicit axioms runs through it unchanged, and the corpus mints sixty of them — which is precisely why a theorem can quietly consume axioms its statement never mentions. The oracle is exact, so every concept of the framework has a decidable or derivable meaning; and the corpus is small enough to audit completely yet rich enough that, in one instrumented week, the framework found and repaired three axiom transcription defects (two by deriving a contradiction), dissolved three phantom walls (of the nine its corpus records), revoked one vacuous theorem and later rehabilitated it by discharge, exposed 53 axioms of invisible debt behind a "hypothesis-free" theorem, and ended with the full 3,909-test suite green (Section 6). This is deliberately the *last mile* of formalization: the gap between a corpus that merely builds and one that is faithful to its source, honest about its debts, and instrumented about its failures — and it is that instrumented mile, not any single theorem, that this paper measures.

Contributions.

1. The first mechanization, to our knowledge, of Bourbaki's own formalism: τ, assemblies and the C1–C62 criteria, as a live, tested corpus (Section 2).

2. Certified failure objects with computed scope, and the E1–E7 taxonomy (seven classes identified, each instantiated; the list is open), every class instantiated by a real, measured case (Section 3).

3. Debt measurement $Ax(D)$, the axioms a derivation D actually consumes — with the measured gap between declared hypotheses and consumed axioms — and an operational trichotomy of open hypotheses whose third branch (independence) we found nowhere else, including a negative result on its tempting syntactic shortcut (Section 3).

4. Mechanized fidelity: infidelity as a *derived* verdict against a pre-existing printed source, two defects detected and repaired this way, the healing itself proved, and the measured cost — two statements that were provable only through the inconsistency (Section 4).

5. A training-free vector detector for twin axioms, calibrated on pairs of known status, and the instance-traced returns of the corpus-guided loop (Section 5).

Section 7 positions each contribution against a systematic, adversarially-conducted related-work sweep; Section 8 states what a single-operator, exact-oracle study can and cannot claim.

2 Background: Bourbaki's Formal System and the Kernel

2.1 A formalism nobody inhabits

In *Théorie des ensembles*, quantifiers do not exist as primitives. They are *abbreviations* built from Hilbert's choice operator. Write R for a relation, x for a letter, and A for an *assembly*: a finite sequence of signs — the four logical signs $\square$ (which marks a position a bound variable used to occupy), τ, $\vee$, $\neg$, the letters, and the specific signs of the theory — of which certain pairs are joined by *links* (E I.14, ll. 11–16). Then

$$(\exists x)R := (\tau_x(R) \mid x)\, R,$$

where $(A \mid x)\, R$ is Bourbaki's notation for the assembly obtained by substituting A for every occurrence of the letter x in R — written $R[A/x]$ elsewhere. Since $\tau_x(R)$ itself contains a full copy of R, unfolding a quantifier copies the formula into itself; nested quantifiers grow multiplicatively per level. The consequences are famous: fully expanded into primitive signs, Bourbaki's numeral 1 occupies 4.5×10^{12} symbols (Mathias), about 2.4×10^{54} in the 1970 edition (Solovay). The links carry the second difficulty: each ties an occurrence of $\square$ — which marks the position a bound variable used to occupy — to the τ that binds it. An assembly is therefore a graph, not a string and not a tree, and has no native syntax in any proof assistant. Finally, since the bound variable disappears into the links, the system has no α-conversion: substitution is trivial by design, which is precisely the property Grimm cites when explaining why Gaia formalizes the *content* of Bourbaki over Coq's own logic and treats Chapter I — assemblies, τ, the deduction criteria — as description only. To our knowledge (Section 7), no prior mechanization inhabits the formalism itself; it is, however, exactly where Bourbaki's metatheory lives: the criteria are theorems *about* theories — among them C62, the licence for every definition by recursion, which the corpus leans on throughout.

2.2 The kernel: inhabiting it anyway

The kernel is written in pure Python, in the LCF style.

Definition 1 (Abbreviated terms and relations)**.** *The sets $\mathcal{T}$ of* terms *and $\mathcal{R}$ of* relations *are defined by mutual induction, exactly as the kernel represents them:*

$$T ::= x \mid s(T_1,\dots,T_n) \mid \tau_x(R) \qquad \textit{(letter, specific sign, choice)}$$
$$R ::= (T = U) \mid (T \in U) \mid \neg R \mid (R \vee S) \mid (\exists x)R \qquad \textit{(the five relation-formers)}$$

where x ranges over letters, s over the specific signs of the theory, and $T, U \in \mathcal{T}$, $R, S \in \mathcal{R}$.

These eight are the constructors of the ABBREVIATED layer, the one the kernel manipulates — and they must be distinguished from Bourbaki's primitives, on pain of drawing the boundary in the wrong place. The book's logical signs are *four*: $\square$, τ, $\vee$, $\neg$. In the list above:

- $(\exists x)R$ **is already an abbreviation** in Bourbaki's sense — it is $(\tau_x(R) \mid x)\, R$ (Section 2.1). The kernel gives it a node of its own because unfolding would mean manipulating assemblies of astronomical size;
- $=$ and $\in$ are not *logical* signs but *specific* ones, relational of weight 2: $=$ belongs to egalitarian theories (E I.38), $\in$ to the theory of sets. They are therefore a special case of $s(T_1,\dots,T_n)$, singled out here only because the kernel treats them separately;
- $\forall$, $\wedge$, $\Rightarrow$, $\Leftrightarrow$ are abbreviations to which the kernel gives no node at all.

The list describes a *representation*, not primitiveness. Writing a sign commits one to the theory that owns it, and this definition borrows two; left unsaid, the abbreviated layer would read as if it were Bourbaki's language.

Definition 2 (Theorem object)**.** *Write $\mathcal{P}_{\text{fin}}(\mathcal{R})$ for the set of finite subsets of $\mathcal{R}$, and let $\mathcal{J}$ be the set of justification labels: the name of the kernel rule applied and,* *for the axiom rule, the theory it was drawn from — which is what makes the debt of Section 3 observable at all. A* theorem *is an element*

$$\theta \in \mathcal{P}_{\mathrm{fin}}(\mathcal{R}) \times \mathcal{R} \times \mathcal{J}.$$

Its three projections are written H, C and j:

- $H(\theta) \subseteq \mathcal{R}$*, finite — the* hypotheses *θ still depends on;*
- $C(\theta) \in \mathcal{R}$ *— the* conclusion *θ asserts;*
- $j(\theta) \in \mathcal{J}$ *— the* justification*, that is, the rule that produced θ.*

The theorem is closed *when $H(\theta) = \emptyset$. Equality is on the first two projections only,*

$$\theta = \theta' \iff H(\theta) = H(\theta') \text{ and } C(\theta) = C(\theta'),$$

so the justification records how a theorem was obtained, not what it asserts.

Example 1 (A theorem, end to end). Take the relation $x \in E$: by Definition 1 it is built by the $\in$ former from two letters. Assuming it gives the theorem

$$\theta_1 : \quad H(\theta_1) = \{\, x \in E \,\}, \qquad C(\theta_1) = (x \in E), \qquad j(\theta_1) = \text{hypothesis},$$

which is *not* closed — it asserts its conclusion only under its own assumption. Discharging that assumption with the deduction rule gives

$$\theta_2 : \quad H(\theta_2) = \emptyset, \qquad C(\theta_2) = \big((x \in E) \Rightarrow (x \in E)\big), \qquad j(\theta_2) = \text{deduction},$$

which is closed. Nothing was proved about E along the way: what the second object records is that the hypothesis has moved from $H(\theta)$ into $C(\theta)$. That move is the whole of *discharge*, and Section 3 turns on the fact that it is the only thing a hypothesis count reflects.

Theorem objects are opaque: they are constructible only through the kernel's rules, and no other code in the corpus may fabricate one.

Bourbaki's own demonstrations use a *single* rule: a demonstration is a sequence in which every relation is an axiom — explicit, or an instance of a scheme — or is detached from two earlier ones, R and $R \Rightarrow S$ (E I.22, ll. 16–25). Everything else he needs, including the deduction criterion and generalization, he establishes as *criteria*: metatheorems about demonstrations, proved once.

Our kernel departs from this on two counts, and both enlarge the trusted base deliberately. It takes the deduction criterion and generalization as *primitive* rules rather than re-deriving them at each use — the code marks them as trusted primitives, not derived — and it carries the hypothesis set H inside the theorem object, where Bourbaki would instead move to a stronger theory. Beyond these, the rules are detachment, the schemes S1–S7, and explicit axioms.

The schemes are not a flat list, and set theory is not the whole story. Bourbaki builds theories in a tower, each level adding its own *implicit* axioms: S1–S4 belong to the *logical* theories (E I.25), S5 to the *quantified* theories (E I.33), S6 and S7 to the *egalitarian* theories (E I.38). Set theory is then the egalitarian theory that adds the membership sign together with its own *explicit* axioms, and the corpus is laid out along that tower rather than around its top floor. This is why the invariant we check counts 22 *explicit* axioms: everything below is inherited as schemes, and a theorem may consume any of it without the count moving — the gap that Section 3 turns into a measurement.

The criteria are stratified the same way, and they are not kernel rules. Each is a metatheorem about demonstrations *in a theory*, proved at the level where it first becomes statable — most of them among the logical theories, generalization (C27) among the quantified ones, and the later ones only once set theory is available. Of the whole series the kernel takes exactly three as primitives: detachment C1, deduction C14, generalization C27. The others are derived, and a good many of them are derived inside the corpus itself, which is what makes "we inhabit C1–C62" a claim about a tested library rather than about a design intention.

Two design decisions make the formalism workable:

Abbreviated trees, expansion as justification. Building the axiom $A1$ of extensionality by full τ-expansion exhausts memory — a `MemoryError` reproducible on demand and journaled, faithful to Bourbaki's own remark on the practical necessity of abbreviations. The kernel therefore represents $\forall/\exists$ as first-order abbreviators at tree level; the τ-expansion remains the *justification* of the rules, not the runtime representation.

What is left unexpanded is the *quantifier abbreviation*, not τ. The choice operator is a primitive term-former of the kernel, governed by Bourbaki's own scheme S7 (E I.38, ll. 22–24): for any relations R and S and any letter x,

$$(\forall x)(R \Leftrightarrow S) \;\Rightarrow\; \tau_x(R) = \tau_x(S).$$

The operator is inhabited throughout the corpus. For a graph f, that is a set of ordered pairs, and any object x, the value $f(x)$ *is* the term $\tau_y((x, y) \in f)$ (E II.13). The term is well formed whatever f and x may be; it denotes what one expects of a value only when f is functional and x lies in its domain. Keeping that gap visible rather than hiding it behind a typing discipline is what makes a proof fail loudly instead of quietly, and Section 3 is largely about what that buys. The projections of a pair, the cardinals, and every canonical witness we build in place of an appeal to choice are τ-terms in the same way. This is where the system departs from assistants that take $\forall/\exists$ as logical primitives and carry no choice operator in the kernel at all. It is also paid for: because $\exists$ is a primitive node rather than a τ-abbreviation at runtime, a few results proved at the τ level — reflexivity of equality among them — are transported into the abbreviated level as trusted primitives. That is the third and last enlargement of the trusted base, and the code marks each one.

The residual cost is real and measured: cardinal terms are τ-terms, constructing $\mathbb{N}$-dependent objects costs minutes per process, printing a cardinal term is never attempted (its expansion is intractable in practice), and one instantiation from variables to closed terms takes a formula from 9,388 to 86,598 characters. This cost is a property of the formalized system, not of the implementation; it is also, as Section 5 shows, the reason caching and vectorization matter.

No α-identification. The kernel does not identify α-variants: two α-equivalent formulas are distinct objects, compared by a dedicated predicate and converted by an explicit certified bridge. This strictness is what makes drift (Section 3) detectable at all — and what makes hard-coded binder names a recurring, measurable trap.

Dedicated theories. Beyond the 22 explicit axioms of the reference theory, the corpus mints small ad-hoc theories: 60 factories at the time of writing, 19 of them parametric. Each comes from Bourbaki's scheme of selection, which licenses naming the set of elements of a given set that satisfy a given relation, and giving that name its own characterizing axiom. The device is legitimate and unavoidable, and it has one consequence nobody measures — the invisible debt of Section 3 — and one genuinely dangerous corner: theorem objects carry no trace of their theory, so the kernel cannot see that a free variable of an ad-hoc axiom is a *constant* of that theory when generalizing.

The corpus at the time of writing: 4,221 collected tests re-deriving its theorems at each run, 2,234 notions anchored to the printed book at page granularity, and a 232-event journal of certified failures — the object of study of this paper.

Its weight is worth stating, because the name of the book invites a wrong picture. The 2,181 notions anchored at the time of the campaign reported in Section 6 distribute over the tower and the chapters it carries as follows (the corpus has since grown to 2,234; the breakdown is a snapshot, not a running total):

§I.1	assemblies, formative constructions	82
§I.2	demonstrations, the criteria, the kernel	133
§I.3	*logical* theories	5
§I.4	*quantified* theories	32
§I.5	*egalitarian* theories	22
Ch. II	sets proper	720
Ch. III	ordered sets, cardinals, integers	1,002
Ch. IV	structures	185

Two readings matter here. First, the set-theoretic level is about a third of the corpus and the largest single part is Chapter III, while the claim of novelty — inhabiting τ, assemblies and the criteria — concerns Chapter I: *Théorie des ensembles* names the book, which erects the whole tower, not one privileged level. Second, the three theory levels look thin only because the corpus is filed by *kind* rather than by floor: the criteria proved in the logical theories are counted under §I.2 with the rest of the criteria, which is where a reader of the repository will find them.

A third reading, and the one that matters most: this table does not say the book is done. It counts what is *formalized and anchored*, not what the book contains. The formalization is partial and uneven, and two figures say so better than a caveat. First, among notions of a kind that *promises a proof* — proposition, theorem, corollary, criterion, lemma — 815 are adjudicated by the kernel and **only 413 are closed, that is 50.7%** (measured 18 August 2026); the rest are proved under honestly declared hypotheses, or not yet. Second, major results of the book remain open in the corpus: Hessenberg's theorem, the well-ordering of cardinals, Cantor's theorem, Euclidean division, and projective and inductive limits.

We insist because the table invites exactly the opposite reading: one sees "§I.1: 82" and understands "§I.1, done". The corpus is large enough to carry this paper's results, and it is not the book. This is the same defect Section 4 treats, applied to a table rather than to a statement: nothing in the repository prevents an exact number from suggesting a false thing.

3 Certified Failure: Definitions, Taxonomy, Objects

Throughout, a *theory* $T = (name, A_T)$ is a name together with a finite set $A_T \subseteq \mathcal{R}$ of relations, its *explicit* axioms. Its implicit axioms — the schemes S1–S7 of the tower of Section 2 — are deliberately absent from the pair: every theory in this corpus stands on the same egalitarian floor, so the schemes are kernel rules available to all of them rather than data carried by each. We write T_0 for the reference theory, so that A_{T_0} is the 22 explicit axioms of the set-theoretic level; the ad-hoc theories minted by selection are further T's, usually with a single axiom of their own, and it is their invisibility to a theorem's statement that the debt below measures. Finally θ is a theorem object in the sense of Definition 2, with projections $H(\theta)$, $C(\theta)$, $j(\theta)$.

We write $\Gamma \vdash R$, for a finite $\Gamma \subseteq \mathcal{R}$, when the kernel can construct a theorem θ with $C(\theta) = R$ and $H(\theta) \subseteq \Gamma$; $A_T \vdash R$ additionally allows the axioms of T to enter through the axiome rule, and $\vdash R$ alone means $\emptyset \vdash R$, a closed theorem. This is a statement about what the kernel *can build*, so every $\vdash$ in this paper is witnessed by an object, never asserted.

The negation $\Gamma \nvdash R$ is not symmetric with it, and the asymmetry matters throughout. $\vdash$ is established by exhibiting one object; $\nvdash$ asserts that *no* object exists, which the kernel cannot witness. Wherever $\nvdash$ appears below it is therefore either a proved refutation ($\Gamma \vdash \neg R$, a stronger claim) or a *declared default* after a bounded search — never a theorem. The trichotomy of Section 3.3 is built on exactly that distinction, and $\vdash$ and $\nvdash$ are metalinguistic notations about the kernel, not signs of any theory it manipulates.

A *goal* is a pair (T, P) with T a theory and $P \in \mathcal{R}$ the relation to be derived in it; we write the goal as P alone whenever the theory is fixed by context, which it is throughout this paper unless stated. A *route* to a goal is one attempted derivation of P, and the *residue* of a route is the set of hypotheses the theorem it reaches still carries — that is, $H(\theta)$ for the θ the route produced. A goal typically has several routes, and the same hypothesis may be open on one and closed on another; that is why the verdicts below are stated of a hypothesis *on a route*, not of a hypothesis outright.

Bourbaki's system has no primitive falsum; we fix the closed witness $\bot := (\emptyset \in \emptyset)$, since $\vdash \neg(\emptyset \in \emptyset)$ is closed from the axiom of the empty set.

3.1 Debt: what a derivation actually assumes

Definition 3 (Consumption and debt)**.** *Call a* derivation *D of a theorem θ the finite sequence of kernel-rule applications that produced it. One of those rules is*

$$\mathsf{axiome}(T, \alpha), \qquad T \textit{ a theory, } \alpha \in A_T,$$

which returns the closed theorem asserting α and refuses any α outside A_T: it is the only way an axiom can enter a derivation, which is what makes the following observable at all. Let

$$Ax(D) := \{(T, \alpha) \mid \textit{the rule } \mathsf{axiome}(T, \alpha) \textit{ occurs in } D\}.$$

The debt *of θ derived by D is*

$$Debt(\theta) := H(\theta) \cup \{\alpha \mid (T, \alpha) \in Ax(D),\, T \neq T_0\},$$

and the real invariant *is $Ax(D) \subseteq \{T_0\} \times A_{T_0}$.*

Both lines are worth reading in words. The first says that what θ actually leans on is what it still assumes, *plus* every axiom its derivation drew from somewhere other than the reference theory — and that only the first half of that is visible in its statement.

The second is the honest form of "the theory has 22 axioms". That familiar invariant is a property of a theory *object*: one counts the axioms of T_0 and gets 22, today and every day, whatever anyone proves. The invariant above is a property of a *derivation*, and it says something much stronger — that this particular proof owed nothing outside the reference theory. A corpus can satisfy the first everywhere and violate the second constantly, and that gap is what the rest of this section measures.

A theorem's hypothesis count records only what *discharge* left behind — the kernel's deduction rule removes a hypothesis by turning it into an implication, and what survives that process is $H(\theta)$. It records nothing about what the derivation had to reach for along the way. The debt does: it keeps the hypotheses and adds every axiom drawn from a theory other than T_0, and nothing in the statement bounds that addition.

Our measured instance is a theorem of the corpus advertised as "closed, 0 hypotheses" — the well-ordering of $\mathbb{N}$. Its derivation was measured to apply the axiome rule **67 times, 53 of those on ad-hoc theories**. Both its statement and the global axiom count, which stayed at 22 throughout, were blind to all 53.

The instrument is an observer placed around the axiome rule: the kernel is not modified, and each time the rule is entered the pair (T, α) is recorded. This makes one failure mode intrinsic and worth stating. Several proof functions in the corpus are memoized; on a second call within the same process the cached theorem is returned and its body never runs, so the rule is never entered and the observer sees an empty $Ax(D)$ for a derivation that does consume axioms. Measured, this undercounts by up to 62%. A measurement is therefore valid only in a fresh process and in first position — and the corpus carries a test that *exhibits* this false negative rather than papering over it.

3.2 Vacuity, drift, phantom walls

Definition 4 (Vacuity). *θ is* vacuous in *a theory T, written $Vac_T(\theta)$, when $A_T \cup H(\theta) \vdash \bot$. Vacuity is thus relative to T: the same theorem can be vacuous in one theory and sound in a stronger one, which is exactly what rehabilitation exploits below.*

A theorem can be perfectly derived and yet worthless: soundness is a property of the derivation, vacuity of the model class. Our measured instance: $0! = 1$ (Def. 2 route) was proved under $\mathcal{H} := (\forall G)(G \in \prod(u, \emptyset) \Rightarrow \mathrm{Gr}(G))$, and $\{\mathcal{H}\} \vdash \bot$ was then derived — the theorem was revoked. Crucially, it was later *rehabilitated unchanged*: after the axiom repair of Section 4, $\mathcal{H}$ became a closed theorem and the identical proof skeleton re-derived with the hypothesis discharged. A vacuous theorem is not a wrong theorem; it is a theorem waiting for its theory.

Definition 5 (Drift). *Let $C^* \in \mathcal{R}$ be the* intended conclusion *of θ: the relation θ is meant to have, written out from the book's statement in a test file that shares no code with the module proving θ. Then*

$$Drift(\theta, C^*) \iff \theta \textit{ builds, is closed, passes its tests, and } C(\theta) \neq C^*$$

as relations — syntactic inequality, not failure.

The independence of C^* is the whole point. A test that compares θ's conclusion against an expression built by the *same* module compares a thing with itself: change the module and both sides move together, and the test stays green while the theorem quietly becomes a different theorem. Drift is thus invisible to the kernel, since nothing is false, and invisible to any test that is not independently anchored.

Measured instance: after an axiom change, one theorem of the corpus still built and remained closed while its conclusion grew from 6,908 to 8,592 characters; its test asserted only closedness and would have stayed green. Two rules follow, and we apply them: targets are rebuilt outside the module and compared syntactically, and hypothesis sets are compared by exact set equality, never by cardinality.

3.3 The trichotomy of open hypotheses

Definition 6 (Trichotomy)**.** *For an open hypothesis h carried by a route over T: h is* dischargeable *iff $A_T \vdash h$;* refutable *iff $A_T \vdash \neg h$ (then every θ carrying h is vacuous);* independent *otherwise — no amount of proof search will close it, and the only legal moves change the theory — as in Figure 3.*

All three classes occurred, measured, within one week: $bo(\leq, \mathbb{N})$ — declared an "irreducible residue" for weeks — is dischargeable (closed, 0 hypotheses); $\mathcal{H}$ above is refutable; the family-value bridge $X_\iota = f(\iota)$ is classified independent of A_{T_0}. The fourth epistemic state, *unknown*, is a measurement debt, not a wall: re-classifying it after each infrastructure fix eliminated nine phantom walls in our corpus.

Proposition 1 (A negative result)**.** *The syntactic criterion "some symbol of h is constrained by no axiom of T_0" is* not *a sufficient test of independence.*

Counterexample, measured: $symboles\_libres(bo(\leq, \mathbb{N})) \neq \emptyset$, yet the hypothesis is provably closed. The criterion is an *indication* (a reinterpretation argument is available) and must be applied only after a prover has failed; applied first, it would have manufactured one more phantom wall.

The asymmetry within the trichotomy is essential. *Dischargeable* and *refutable* are witnessed by kernel objects; *independent* is not, and cannot be: it is the *absence* of both witnesses. The classifier therefore returns *independent* only after a prover has failed and the free-symbol indication holds — an ordering of evidence, not a proof. Independence and *unknown* are distinguished operationally, by the strength of the prover that failed, never certified. This is the one branch of the trichotomy whose verdict is defeasible, and the corpus records it as such.

3.4 Failure objects and their validity conditions

Definition 7 (Certified failure)**.** *$Echec = (P, \kappa, c, backtrack, scope)$ — we keep the repository's French names, Echec (failure) and Mur (wall), to match the code. Here P is the* goal *in the sense fixed above — a relation of $\mathcal{R}$ to be derived in a theory — not the route and not the theorem it reached; $\kappa \in \mathcal{K}$ its failure class, where $\mathcal{K}$ is the set of classes we have* identified *and $|\mathcal{K}| \geq 7$ (see below); c a kernel object certifying that class, backtrack the stated change that avoids falling into it again, and scope the* computed *set of goals the failure condemns.* verifier *accepts iff c matches κ: for $E2$ (vacuity), $C(c) = \bot$ and $H(c) \subseteq$ the route's residue; for $E5$ (phantom), c is closed and $C(c)$* is *the declared residue; etc. A failure whose certificate fails verification is* rejected from the corpus*.*

Definition 8 (Wall)**.** *$Mur = (condition, predicate, c)$: while condition is unrepaired, every goal satisfying predicate is doomed, and predicate* computes *the scope rather than asserting it.*

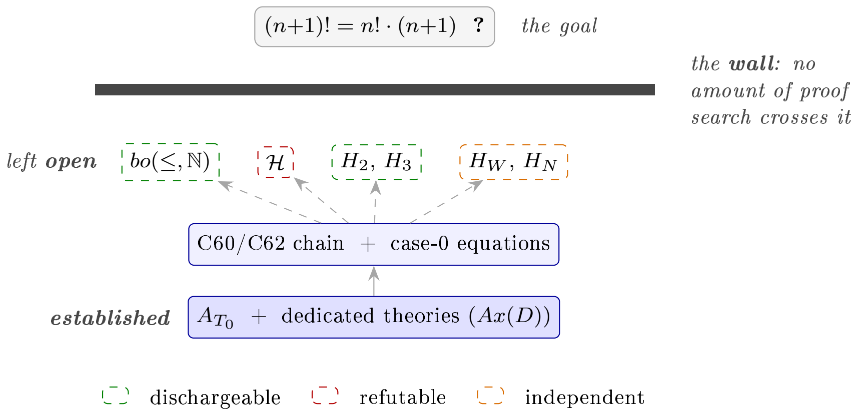

Figure 1: A real wall: the factorial front's last open equation. **Read from the bottom up**: at the bottom what is established (solid boxes — axioms, then kernel theorems), above it the four hypotheses the derivation chain leaves *open* (dashed boxes), and the bar is the wall standing between them and the goal. The colour of an open box is its trichotomy verdict, given in the key: only the orange one — *independent* — is a wall in the strict sense, the other two close. The four boxes correspond one-to-one to the rows of Table 1.

open hypothesis	met on	verdict	outcome
$bo(\leq, \mathbb{N})$	the C62-existence route	dischargeable	declared irreducible, later proved closed
$\mathcal{H}$	the case-0 route	refutable	source of the vacuity; theorem revoked
H_2, H_3	the recursion step	dischargeable	fell with the axiom repair of Sec. 4
H_W, H_N	the recursion step	independent	no search closes them; only the theory can change

Table 1: The front's frontier, aggregated across its routes rather than taken from a single journal snapshot. The recursion step's measured residue was $\{n{\in}\mathbb{N}, H_2, H_3, H_W, H_N\}$, and the trichotomy split it. Each row is a measured case, and each failure is recorded as a certified object.

Measured instance: the refutation witness behind $\mathcal{H}$ hard-codes the empty index, so the wall's predicate provably reaches only products over a literally empty index — H2 (index $I \cup \{j\}$) and H3 (variable index) were *proved out of reach*, and exactly one file was revoked. Figure 1 shows such a wall in place, with the verdict of each hypothesis it leaves open.

The taxonomy is open, and cannot be closed by the same means as the rest. We write $\mathcal{K}$ for the set of failure classes and $|\mathcal{K}| \geq 7$ rather than $\kappa \in \{E1, \ldots, E7\}$, because the second notation asserts something we have not established. Seven classes were identified and each was instantiated by a real case; that a failure exists which fits none of them is neither excluded nor excludable here. The classes were found by looking at failures, not derived from a principle, so the catalogue grows by encounter: a failure matching no κ becomes $E8$.

class	definition	valid certificate	measured instance
E1 drift	$C \neq C^*$	syntactic comparison	*membre_but* (6,908 → 8,592 chars)
E2 vacuity	$\{h\} \vdash \perp$	theorem with $C = \perp$	$\mathcal{H}$ / revoked $0! = 1$
E3 dead end	no continuation closes	prunes extensions	the "$n-1$" route (opaque difference term), ev. 84–85
E4 debt	$Ax(D) \not\subseteq \{T_0\} \times A_{T_0}$	fresh-process measure	*n_bien_ordonne*: 53 foreign axioms
E5 phantom	residue already provable	closed theorem = residue	$bo(\leq, \mathbb{N})$, 9 cases total
E6 infidelity	$A_{T_0} \cup \Phi \vdash \perp$	closed derivation + source anchor	two axiom defects (Sec. 4)
E7 measurement	the probe lied	counter-measurement	name-guessing probe (false negative)

Table 2: The failure taxonomy *as far as we have identified it.* Every class listed is instantiated by a real, kernel-checked case from the instrumented week. The list is **open**: nothing here shows that seven are all there are.

The asymmetry is the same one this paper meets twice elsewhere. Infidelity is semi-decidable — one can always *find* one, never certify their absence (Section 4); novelty likewise (Section 7). Exhaustiveness of $\mathcal{K}$ is of that family: exhibiting an eighth class is a finite act, proving there is none is not. We therefore report a catalogue with a lower bound, not a type with seven inhabitants, and we would rather the reader find an $E8$ than believe the list closed because we typeset it with braces.

4 Mechanized Fidelity: When the Corpus Refutes the Book

4.1 The anchoring discipline and the infidelity verdict

Every formalized notion carries a machine-readable anchor `@livre Ch.X §s.ss Type.n | E X.pp L.a-b | PDF p.n` tying it to the printed source (2,234 anchored notions at the time of writing). The anchor has two fields of different reliability, and we separate them because we measured them.

The page anchor is sound; the line interval is not uniformly so. The page field is what carries the coverage results below: zero non-conforming anchors, and five parts of the book reported complete over their intervals. The line interval is weaker, and we discovered this by auditing our own anchors against the book. On page E II.22 — 36 printed lines, counted by hand — Definition 1 stands at lines 15–18 and Definition 2 at 27–30, while their anchors read `L.31-36` and `L.49-53`; the second exceeds the page. Line 53 of our LaTeX *transcription* of that page, however, is exactly the header of Definition 2: those two anchors index the transcription, not the book. On page E II.7 the opposite holds — the anchor tracks the printed line with a small offset and does not match the transcription at all. At least two conventions coexist in the corpus.

Measured over the 1,992 anchors carrying a line interval: **12.2% end beyond line 36**, which is about a full page of this edition, so they cannot be printed-line references; the maximum is 63. The bulk of the distribution has the shape of a printed page, with a tail that does not.

We therefore claim traceability *at page granularity*, which is what the coverage instruments actually consume, and we report the intra-page line intervals as a *heuristic* for locating unformalized material rather than as a certified measure. This is a weaker claim than the one this paper made in draft, and the reason to record it here is that it is the paper's own thesis turned on the paper: no test in a kernel-verified repository catches an anchor that misdescribes the book, and ours went unexamined until we opened the pages. Let Φ be the transcription of the book's numbered statements. Fidelity itself — "$\varphi(b)$ *means* b", for a statement b of the book and its formalization $\varphi(b)$ — relates text to formulas and is not formalizable; its negation is:

$$\mathit{Infid} \iff A_{T_0} \cup \Phi \vdash \bot.$$

Infid is semi-decidable: one can always *find* an infidelity, never certify its absence — which is why our audit is a standing process rather than a coverage percentage. Two consequences of the discipline are mechanizable for free: two anchors citing the same sentence at different lines expose an error by mere comparison (two real cases found and fixed); and a copied line number propagates — one wrong value was found replicated verbatim across five sites.

Recent project-scale autoformalization also records span-level provenance; the state of the art, M2F [44], then enforces statement faithfulness "by a provenance-linked *manual audit*" (p. 8). Our departure is precisely there: the infidelity verdict is *derived*. Both defects below were established by proving a contradiction in the kernel, and both repairs were proved effective by the impossibility of re-deriving it.

A third defect, found the same week, seeded the method. The intersection axiom had been transcribed from the book's *Summary of Results* (E R.19), whose universe of families of *subsets* makes $\bigcap_\emptyset = E$ legitimate, rather than from E II.22, whose Definition 2 requires $I \neq \emptyset$ (and whose footnote warns that the unrestricted intersection is not collectivizing). The reference theory itself was therefore inconsistent. The defect was localized by machine proof, its blast radius computed at 34 sites, and repaired by restoring the selection form, holding the axiom count at 22; the incident was closed, suite green, before any debt measurement reported in this paper was taken. The two cases below were then found *by* the discipline that this first incident instituted.

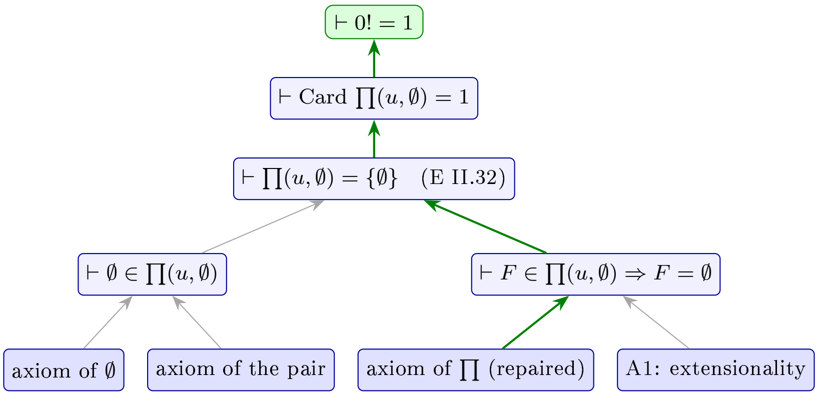

Figure 2: The real derivation of $0! = 1$ (Def. 2 route) as a graph inside the theory box. **How to read**: every box is a kernel object — bottom row, the axioms consumed; above them, derived theorems; the green-filled box is the goal. An arrow from A to B means "A is a premise of the kernel step that derives B". **Thick green arrows** mark the branch that was *underivable* while the product axiom lacked its conjunct; **thin gray arrows** are steps that never depended on the defect. The whole derivation is the repair's dividend.

4.2 Case 1: an axiom lost a conjunct

The product-of-a-family axiom had lost the book's bound on the graphs it admits: for a family $(X_\iota)_{\iota \in I}$ of sets indexed by I, a member F of the product must satisfy $F \subset I \times \bigcup_{\iota \in I} X_\iota$ (its sibling, the exponentiation axiom, had kept its own). The corpus then proved, at *zero* hypotheses and for any family u, both $\{\emptyset\} \in \prod(u, \emptyset)$ and $\neg(\prod(u, \emptyset) = \{\emptyset\})$ — while E II.32 states that the empty product has exactly one element. The corpus refuted the book. Soundness was never at issue: the kernel faithfully derived from the axiom it was given; the axiom was not the book's. The repair restored the conjunct *in head position* (an empirical choice: 18 broken tests against 33 for tail position), kept the axiom count at 22 (replacement, not addition), and paid immediately: two hypotheses carried for weeks became theorems, and the vacuous $0! = 1$ of Section 3 was rehabilitated by discharge. Figure 2 shows that rehabilitated derivation, with the branch the defect had blocked marked in green.

4.3 Case 2: a term lost a parameter

The term $seg(E, x)$ — the initial segment determined by x — did not carry its order relation: two different orders produced the *same* term, each with its own characterizing axiom, both accepted at zero hypotheses under one theory name. From two concrete orders on a two-element set, $\vdash \emptyset \in \emptyset$ follows in six kernel steps. A mechanical AST pass then generalized the finding: **21 term constructors drop at least one parameter, 11 of them the order**, with a second contradiction derived for closed intervals.

Proposition 2 (Guards do not repair lost parameters)**.** *Adding the hypothesis "R is an order on E" to the characterizing axiom leaves the contradiction derivable: two* different *orders on the same E both satisfy the guard. A lost* condition *can be restored by a guard (Case 1); a lost* parameter *cannot — no condition repairs a term that has no slot for it.*

The structural cure: the term carries its graph — $seg(G, E, x)$, for a graph G, a set E and a point x — and the axiom is $\forall$-closed over G as well. The four α-distinct axiom instances collapse into one closed formula; the theory factory loses its parameter; and the contradiction becomes *structurally underivable* — there no longer exists a mechanism for minting two incompatible axioms about one term. As a side effect, the closure removes the free variable that had let the kernel generalize over a *constant* of the ad-hoc theory (a boundary condition of Bourbaki's C27 that the kernel cannot check, for the same root cause as the invisible debt of Section 3: theorem objects carry no trace of their theory). The full 3,909-test suite is green under the repaired axiom.

4.4 The price of truth: load-bearing contradictions

Repair had a cost, and naming it is part of the method. Two intermediate theorems had eliminated Zermelo's $\exists R_o$ *because* the segment term did not carry its order — their claimed "R_o-independence" was an artifact of the defect. Post-repair, the kernel rightly refuses the generalization; the statements were restated in R_o-closed form, and the loss is named: *Zermelo alone no longer suffices* for these two reductions — the well-order and the realisation of segments by it must be assumed together. The stronger statements were provable only through the inconsistency. This is the measured dual of vacuity: vacuity is a theorem born dead under a refutable hypothesis; here a theorem *dies with* the defect that carried it. An incoherence audit must therefore look both ways: for what the defect makes spuriously refutable, and for what it makes provable-but-illegitimate. The local documentation had claimed the independence in good faith — it was the ambient theory that lied.

5 The Guidance Loop and the Vector Layer

5.1 The guidance equation

A bare policy proposes $\pi(a \mid s)$; plugged into the corpus it proposes $\pi(a \mid s, M(s))$, where $M(s)$ retrieves the walls whose predicate covers s, the remedy couples whose recorded state resembles s, and the applicable lessons. $M(s)$ acts on direction three ways: it *forbids* (certified walls prune moves toward doomed routes), *suggests* (a remedy couple says: at a state like this, that move failed and this one worked), and *orders* (premises closest to the goal come first). Feedback closes at three timescales:

per move, the kernel accepts or rejects — hallucinated progress does not exist; per route, the frontier $Ouv(D)$ and the trichotomy classes of its open hypotheses give a learning-free value estimate (a route carrying an independent-looking hypothesis is doomed at any effort); per campaign, closed book problems and total debt.

Returns are documented instance by instance rather than averaged: a revoked proof skeleton, kept with its vacuity certificate, re-derived unchanged once the repaired axiom made its hypothesis dischargeable — a full reconstruction avoided; a certified dead-end route (an opaque subtraction term) cost us once and never again, the next agent taking the recommended $succ(k)$ form directly; nine phantom walls produced the standing rule *re-measure before believing*, after which no session was lost to a nonexistent wall. The loop's failures are journaled with the same discipline (a 21/21-wrong triage prognosis; a budget overrun that produced the project's frugality protocol), and the validity caveat is stated in Section 8: these are traced instances, not a controlled effect.

5.2 The vector layer and the twin-axiom detector

Formulas are labeled trees. Writing $\ell_k(v)$ for the label of a node v after k rounds and h for a hash, the Weisfeiler-Leman refinement

$$\ell_{k+1}(v) = h\big(\ell_k(v), \{\!\{\ell_k(u) : u \text{ child of } v\}\!\}\big)$$

hashed into d=512 buckets over $k \leq 3$ gives a deterministic, training-free embedding $\varphi(t)$ of a term or formula t, compared by cosine similarity — computed by walking the abbreviated tree, never by printing it. Two design lessons were learned the expensive way, and measured: a label that drops the symbol name collapses distinct axioms (cos = 1.0000 spuriously), and a walker unaware of one child attribute stops at depth one; the corrective rule is that *feature maps are validated on pairs of known similarity*, the mutation-testing discipline applied to representations. Calibration ships as tests. The incoherent *seg* axiom pair measures 0.9810 — reconstructed as a fixture, the defect itself having since been repaired, so that the detector keeps a positive case to be tested against; two axioms of one family measure 0.8758 and a genuinely unrelated pair 0.5415, so $\theta = 0.90$ separates with margin on both sides.

The detector then flags

$$Twin(\alpha, \beta) \iff \tau(\alpha) \cap \tau(\beta) \neq \emptyset \;\wedge\; sim(\alpha, \beta) \geq \theta_{\text{twin}},$$

a shared characterized term and similar bodies, evaluated on the pairs already retained at $\theta = 0.90$. There are two thresholds, not one: θ selects what is worth looking at, $\theta_{\text{twin}} = 0.95$ pronounces. The conjunction matters: the sweep's highest similarity — a full 1.0000 — holds between two copies of a single formula, an ad-hoc theory still carrying the axiom that the intersection repair had since promoted into A_{T_0}. Its definiendum belongs to T_0's own vocabulary and is therefore filtered out of τ, no term is shared, and no alarm is raised — though the score alone would have raised the loudest one in the corpus.

Defining $\tau(\alpha)$ took one further refinement, and a false positive is what produced it. A first criterion took every symbol proper to an axiom, and so paired the axiom *defining* the integer interval with one that merely *uses* that interval as a bound. Two axioms that mention a term are not in conflict; only two that define it are. The characterized term is therefore the definiendum — the left member of the axiom's defining equivalence — and nothing else. The distinction is not a detail of implementation: it is what separates a similarity score from a verdict, and it was invisible until the detector produced an alarm we could check by hand.

The sweep vectorizes 65 axioms of 41 theory factories — 43 axioms of 40 dedicated theories, plus the 22 of T_0 itself, scanned as controls — in a few seconds, returns 34 pairs above θ and *exactly one* twin (two theories characterizing one term h_iso_max at 0.9911, under audit); 19 parametric factories are skipped and reported as such — they are the risk class itself, a parametric factory being precisely the mechanism that made *seg* incoherent. The incoherence that had cost a 95-minute investigation is now a standing conformity query.

6 Case Study: One Instrumented Week

We report one instrumented week (July 26 – August 2, 2026) during which every mechanism of Sections 3–5 was exercised on live mathematics — the factorial chapter of *Théorie des ensembles* and its upstream machinery.

Three episodes carry the argument. *(i) The full repair cycle, twice.* Both axiom defects ran the complete loop — detect by derivation, localize by measured blast radius, repair structurally, prove the contradiction underivable, re-run the entire suite — demonstrating that an inconsistent ambient theory is a measurable, repairable event rather than a catastrophe. *(ii) Revocation and rehabilitation.* The vacuous $0! = 1$ was revoked with its certificate kept; the repair turned its refutable hypothesis into a theorem; the identical skeleton then closed without residue and survived an adversarial pass (fourteen genuine mutants killed, none by exception). Failure objects are not write-offs; they are deferred assets. *(iii) The price of truth.* The same repair weakened two acquired statements that had leaned on the defect — measured, named, and retained as the strongest evidence that the contradiction was live.

Operationally, the week also measured its own envelope: the reference photo of the full suite ran in 2 h 43 wall-clock at four workers; long runs are killed by the environment near the 50-minute mark, so suites are sharded into seven zones with results persisted incrementally; and an early overrun of agent budget produced the frugality protocol under which the related-work sweep of Section 7 later returned 69 triaged references for 237k tokens where the week's first fan-out had burned 266k for zero results. The harness, too, is part of the experiment.

axiom defects found & repaired	**3** (one by machine proof of the theory's inconsistency, two by derived contradiction), all repaired with the count held at 22, healing proved by non-derivability
documentation claims refuted	∼**78** on the first audit day; a directors' triage prognosis later measured wrong **21/21**
phantom walls dissolved	**3** this week — nine in the corpus overall — declared blockers measured dischargeable or already solved
vacuous theorem	**1** revoked (refutable hypothesis), later *rehabilitated unchanged* by discharge
weakened statements	**2**, provable only through the inconsistency; loss named (dual of vacuity, Sec. 4)
invisible debt measured	4 capstones: **53** / **57** / **9** / **82** foreign axiom instances behind declared hypothesis counts of 0 / 10 / 4 / 7
constructors dropping parameters	**21** flagged by one AST pass; 11 drop the order; 2 contradictions derived
test suite	3,850 → 3,909 collected; three full green runs under successive repairs, final **3,909** / **3,909**
certified event journal	**95** entries at week's end (37 predate the week), each re-parsed and, where applicable, kernel-checked
detector economics	investigation ≈ 95 min → query seconds

Table 3: The instrumented week in numbers. Every figure traces to a journaled, dated event; test totals rose across the three recorded green runs — no suite was silenced to get green.

Reproducibility. All measurements were taken on one Windows 11 machine: Python 3.13, pytest 9.0.3 with `pytest-xdist`, `PYTHONIOENCODING=utf-8`, suites sharded by zone. The test counts quoted in the case study above are those of the instrumented week and are left as measured then; for the record, the suite rerun in full on 20 August 2026 reports **4,221 passed in 2 h 03** at twelve workers with `-dist loadfile`. We report both rather than retrofitting the narrative with a later number. The debt instrument is valid only on the first measurement of a fresh process (Section 3); the twin scan ran with $K = 3$, $d = 512$, $\theta = 0.90$, $\theta_{\mathrm{twin}} = 0.95$, and is reproduced by `python outils_ia/vecteurs/scan_jumeaux.py` — its embedding hashes with `blake2b` rather than Python's `hash`, whose per-process randomization would make every cosine reported here irreproducible. The corpus, the journal, and every script cited here ship in the repository snapshot referenced on the title page.

6.1 Second case study: an open statement under instrumentation

Two days after the week reported above (August 5–6), the same instrumentation was turned toward an *open* statement: the Goldbach conjecture, written over the corpus's constructed arithmetic. We report here only what bears on this paper's theses — that statement defects are *proved* and that debt is *measured.* The reductions the episode produced, and the measurement of how much arithmetic they contain, are the subject of a companion paper; the discovery organs it exercised belong to another still.

(i) Two statement defects, proved in the kernel. The first rendering of the conjecture read "n even and $n \neq 2$". Its antecedent is satisfied at $n = 0$ — proved, closed: $\vdash$ even(0) (witness $k := 0$) and $\vdash \neg(0 = 2)$ — so the statement asserted that *zero is the sum of two primes.* Repaired, it was still wrong: "even" constrains n to be a *cardinal*, not an integer, and the antecedent is also satisfied by $n := \mathbb{N} + \mathbb{N}$, which is infinite — likewise proved, the equality with the old antecedent instantiated being checked formula against formula. The same fault — a quantifier ranging over sets while believed to range over integers — had been committed the day before on primality's $(\forall d)$: it is a *pattern*, now on record.

Notably, the *proved* (bounded) variant of the statement carried the finiteness guard from its first draft, because its proof demanded it; only the *unproved* statement had drifted. **Proving forces hypotheses to be named; stating forces nothing.** We regard this as the most transferable observation of the episode, and it is why the fidelity audit of Section 4 targets statements that no proof is currently exercising.

(ii) Debt, in three numbers. "Zero residual hypotheses" and "the 22-axiom invariant holds" read together as "depends only on the 22" — false in the measured sense, since axioms enter through the axiom rule and vanish from the sequent (Section 3). Measured in a fresh process on the bounded n-form: **0** residual hypotheses; **14** of the 22 reference axioms consumed; **53** foreign formulas — one single rule, criterion C54 (the graph of a term-defined function, a theorem in Bourbaki, a dedicated-theory axiom here), instantiated 52 times, plus one Knaster–Tarski instance. An honest result is announced with all three numbers; the third says what remains to be derived.

On the episode's main equivalence the same measurement returns **0**; **14**; **73** — spread over *21* dedicated theories, the well-ordering machinery (Zorn, Zermelo, Bourbaki–Witt) entering through the "a lower bound of a finite is finite" lemma. Removing a bound is paid for in well-ordering: the jump from 2 to 21 theories *measures* what generality costs. The first measurement returned zero in one second — the theorem came out of a cache, and the instrument is blind to what is not built: the fresh-process rule, written for other measurements, applies to itself.

What is deferred. The episode ran four days and left the axiom count at 22. Its mathematical output — a chain of certified reductions collapsing the conjecture onto a single object, and the measurement, by parameterizing the primality predicate, that those reductions contain no arithmetic at all — is reported separately, as is the work of the discovery organs it exercised. None of it comes close to the conjecture. What belongs here is the pair of results above: an open statement is an unusually good place to catch a fidelity defect, precisely because nothing is proving it.

7 Related Work

This section condenses a systematic, adversarial sweep (four independent search tracks, 57 queries, 69 references triaged, the five closest threats read in full); the sweep's method and complete tables are in the repository.

Closest systems. Goedel-Architect [8] is the nearest neighbor: blueprint graphs whose failing nodes carry one of *two* diagnoses — statement-wrong, backed by a compiler-verified counterexample, or proof-too-hard, closed by a structured "forfeit". Our departures are threefold: their forfeits record where the prover *believes* the gap lies (unverified prose) where our certificates are kernel objects with computed scope; their loop lives per run where our corpus persists and returns are traced instance by instance; and their diagnosis is binary where our trichotomy has a third branch. AutoformBot [33], the system behind the ATLAS library, publishes the loop's *form* — persistent trace analyzers maintain "skill guides" that workers must read before retrying — and a deliberately non-transitive success criterion; our claims accordingly rest not on the loop but on the certification of its links and on per-derivation debt. Self-modifying prover agents [23] co-evolve with a benchmark score rather than a certified journal.

Failure data and proof repair. REPLica [35] captured human failed attempts and cancel-repair couples at constant state — the format our remedy events adopt. The program-verification lineage turns failed proofs into structured artifacts: a counterexample-derived executable test [13] with a three-cause diagnosis taxonomy [31]. Repair is now an industry: transformation-based [36], dataset-scale [34], error-message-conditioned [10, 41], blueprint-local [17]. Error taxonomies exist for human attempts [2], version incompatibilities [25], and benchmark defects [5] — the latter the only other work we found that *certifies* the defects it reports. None of these certifies a search failure in the kernel, computes its invalidation scope, or preserves it once no repair exists.

Faithfulness and Bourbaki. Span-level provenance to source texts is now standard at project scale [44, 6], and faithfulness certification is an active 2026 topic [28, 43]; the decisive difference, argued in Section 4, is that the state of the art's verdict is estimated or manual where ours is derived. Proof auditing [1] and inconsistency detection in large FOL bases [37] pursue soundness of a base for itself, without a source text or a certified healing. On the formalism: Gaia [11] formalizes Bourbaki's *content* over Coq and leaves Chapter I described but uninhabited; the size pathology is classical [26]; modern LCF kernels exist in Python [47] and over standard set theory [12], so the vehicle is not the novelty — the inhabited formalism is.

Theory repair, revision, formation. Automated theory repair is an established program [21, 22, 39], grounded in AGM revision [3]; its diagnoses are heuristic and its cures validated empirically, not by derivability change in a kernel. Computational Lakatos exists [30] and theory exploration discovers true lemmas [15]; both discard their failures. Disproof is industrialized, from Nitpick [7] to LLM-scale counterexample generation [24] — the refutable branch of our trichotomy; we measured that the latter never mentions a third status, and the field's 2026 survey [46] organizes the area along axes orthogonal to logical status.

Guidance, embeddings, environments, memory. Structural formula features for large theories are a decade old [16], with learned symbol-invariant embeddings [29, 14] guiding saturation and premise selection [4, 27]; our detector shares the invariance goal but is training-free (exact WL [38]) and solves a different problem — inter-theory coincidence of characterizing axioms, a task we found unaddressed. Kernel-as-environment is mature [45, 19, 32]; agent skill memories validated by execution [42] and wake-sleep library learning [9] are the loop's ancestors outside mathematics. Dependency structure of libraries is analyzed at scale [20, 33], topologically; we found no named metric of per-derivation axiom debt. The deployment frame itself — forward-deployed engineering and context/harness engineering — has just acquired a taxonomy and quality criteria [18, 40]; this paper can be read as an instrumented, verifier-backed instance of that practice.

Novelty, honestly stated. Novelty claims are semi-decidable: one can prove non-novelty by exhibiting the paper, never novelty. Our claim is therefore methodological: after a documented adversarial sweep, no counterexample was found to any of the following — kernel-certified failure objects with computed scope; an operational third (independence) branch; a mechanized, derivation-based infidelity verdict against a pre-existing printed source; a named per-derivation axiom-debt measure; training-free twin-axiom detection; an inhabited Chapter I.

8 Limitations and Threats to Validity

One mind. The system was designed, operated, and audited by a single author-operator (with LLM agents under direction). Adversarial verification passes are institutionalized precisely because the director is not trusted: one triage prognosis was later measured wrong on all 21 of its predictions, and that episode is journaled with the same discipline as the successes. Still, independent replication is the real test, and the corpus is published to make it possible.

Observational returns. The guidance loop's returns (Section 5) are traced instances, not a controlled effect: the week cannot be replayed without its corpus to measure a counterfactual. What survives this caveat: each instance is dated and journaled on both ends (the failure and the reuse), several gains are mechanical ratios independent of any counterfactual (95 minutes to a few seconds; a declared multi-session blocker measured solved in 4.7 seconds), and the loop's failures are recorded with equal discipline.

Scale. 232 certified events, 8 debt measurements, 60 dedicated theory factories, one book. Every claim in this paper is an existence or instance claim, never a statistical one. The debt instrument is valid only on the first call of a fresh process (measured -62% undercount otherwise); the full suite runs in 2 h 03 at twelve workers (2 h 43 at the four used during the week reported here) and the environment kills long runs near 50 minutes, which bounds experimental turnaround and dictated the sharding discipline.

Known unmeasured areas. Assertion quality of tests outside the files we edited (a drift could hide behind a closedness-only test elsewhere); the 19 parametric theory factories not yet coverable by the twin detector — the risk class itself; the downstream sufficiency of the two deliberately weakened statements; the C27 boundary condition in the kernel, documented as design debt and left untouched by project rule; one named fidelity gap (sup vs maximum in C63's $M(u)$) assumed and confined. Novelty rests on a documented adversarial sweep and is semi-decidable by nature; the five closest threats were verified against their full PDFs, and every remaining author list against its arXiv or publisher page.

Generalizability. Everything here enjoys an exact oracle. The framework's transfer to noisy-oracle domains — where certificates become statistical — is plausible but untested; we claim the zero-noise limit, not the general case.

9 Conclusion: Toward Theory-Creating AI

The goal this substrate serves is an AI that *creates theories* to solve problems. Deriving requirements from that goal — theories as cheap first-class data; an exact, unfakeable verifier; traceable debt; invented objects as terms; a corpus of *process*, failures included; metatheory available inside the system; a supply of (T, P) pairs with a difficulty gradient; local composable moves; identity across revision; transport between theories — yields a list we keep explicitly *open*: it was itself proven

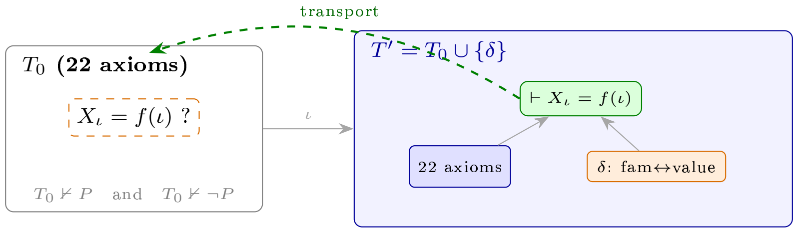

Figure 3: The move that only a theory-as-data substrate permits. **How to read**: each large rounded rectangle is a *theory* (left, the reference T_0; right, its extension T'); inside, blue boxes are axioms, the **orange** box the added definition δ (and, at left, the orange hypothesis it answers), the green box the goal now provable; the **dashed green arrow** is the transport of the result back into T_0's language. When a hypothesis is *independent* (here the family-value bridge, independent of A_{T_0}), no search closes it — the only legal move extends the theory with a definition δ, proves the image, and transports the result back.

incomplete once, by the same adversarial hunts that audit everything else here, and completeness of such a list is exactly as semi-decidable as fidelity. The system described in this paper satisfies the core of it, and the instrumented week is the existence proof that the hard part — the full cycle *detect by derivation, localize by measured scope, repair structurally, prove the healing, name the cost* — runs end to end on live mathematics, twice.

Read abstractly, the system is experimental inquiry in the zero-noise limit: falsification is a derivation, theory revision is an executable operation with a certified outcome, test severity is a mutation count, and the economy of measurement is a budget with journaled overruns — Popper, Lakatos, Mayo, and AGM as running code rather than philosophy of science. That limit is what makes the corpus valuable as training data: every label is exact.

The roadmap follows from the measurements. Replay instrumentation turns the existing 4,221-test corpus into 10^5–10^6 certified (state, action, verdict) tuples without formalizing a single new page — the data factory is the same infrastructure as the term cache that removes the dominant reconstruction cost. An environment API ($\mathit{reset}(T, P)$ / step) exposes the loop to any policy: frontier models are rented and replaceable; the guidance memory $M(s)$ and its certified labels are what accumulate and what no existing proof corpus provides. The twin detector's second version reads the characterized term from the axiom's shape and covers the parametric factories. And the strategic frontier is Bourbaki's Chapter IV — structures and transport — because that is where a theory created for one problem learns to pay off in another.

A proof assistant certifies what we know. A certified failure corpus records how we came to know it — the walls, the repairs, the prices paid — and for a machine that must learn to *create* theories rather than search within one, that is the scarcer commodity.

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